Day 10. Gradient descent for logistic regression

Gradient descent implementation

we are going to use gradient descent to find the best w and b again here in classification.

Gradient descent

cost

$$J(\vec{w}, b) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log\left(f_{\vec{w},b}(\vec{x}^{(i)})\right) + (1-y^{(i)})\log\left(1-f_{\vec{w},b}(\vec{x}^{(i)})\right)\right]$$

repeat {

j=1...n

$$w_j = w_j - \alpha \frac{\partial}{\partial w_j} J(\vec{w}, b) \qquad \frac{\partial}{\partial w_j} J(\vec{w}, b) = \frac{1}{m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)x_j^{(i)}$$

$$b = b - \alpha \frac{\partial}{\partial b} J(\vec{w}, b) \qquad \frac{\partial}{\partial b} J(\vec{w}, b) = \frac{1}{m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)$$

}

Gradient descent for logistic regression

repeat {

looks like linear regression!

$$w_j = w_j - \alpha\left[\frac{1}{m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)x_j^{(i)}\right]$$

$$b = b - \alpha\left[\frac{1}{m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)\right]$$

} simultaneous updates

Same concepts:

- Monitor gradient descent (learning curve)
- Vectorized implementation
- Feature scaling

Linear regression $f_{\vec{w},b}(\vec{x}) = \vec{w}\cdot\vec{x} + b$

Logistic regression $f_{\vec{w},b}(\vec{x}) = \dfrac{1}{1+e^{-(\vec{w}\cdot\vec{x}+b)}}$

Question

Why the partial derirative functions are the same as the ones in the linear regression?

it is quite weird?

$$g(z) = sigmoid(z) = \frac{1}{1+e^{-z}}$$

$$f(x) = \vec{w} \cdot \vec{x} + b$$

$$f_{\vec{w},b}(x) = \frac{1}{1+e^{-\vec{w}\cdot\vec{x}-b}}$$

$$L(f_{\vec{w},b}(x^{(i)}), y^{(i)}) = -y^{(i)} \log(f_{\vec{w},b}(x^{(i)})) - (1-y^{(i)}) \log(1-f_{\vec{w},b}(x^{(i)}))$$

$$J(\vec{w}, b) = \frac{1}{m} \sum_{i=1}^{m} L(f_{\vec{w},b}(x^{(i)}), y^{(i)})$$

to get $\frac{\partial J(\vec{w},b)}{\partial w_j^{(i)}}$, we can calculate $\frac{\partial L}{\partial w_j}$ first.
then use the chain rule we can get the final result.

$$\frac{\partial}{\partial w_j} J(\vec{w}, b) = \frac{1}{m} \sum_{i=1}^{n} (f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}) x_j^{(i)}$$

the complex terms will be elliminated in the chain rule process, and the form of the final result will be the same with the one in linear regression.